

Stress Relaxation of Spring Materials

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Abstract

Stress relaxation behavior is presented for five spacecraft spring alloys. Data were obtained as a function of initial stress for 0.51 and 1.27 mm (0.02 and 0.05 in) wires in uniaxial tension at ambient temperature. Elgiloy and 17-7 PH were the most resistant, followed by 302 stainless steel and music wire. Be-Cu alloy 25 was the least resistant. An analytical technique to predict stress relaxation of springs, based on the uniaxial wire data, is presented.

Introduction

Deployable space hardware, such as solar panels and booms that depend on spring energy for deployment forces, may be compromised by loss of energy in the spring due to stress relaxation during long-term storage. As a result, either a pre-storage or post-storage analysis is required whenever critical hardware having stressed springs is placed in long-term storage. A major obstacle to these analyses is the lack of information available in the literature for predicting stress relaxation of the high-strength alloys used in the manufacture of spacecraft and satellite springs.^{1,2} Much of the stress relaxation data available are for short times, typically up to 100 h, relative to the multi-year storage durations anticipated for many of the current space systems. This often leads to an undesirable circumstance in which extrapolation of sparse data over several orders of magnitude is required for the analysis.

To aid this situation, a test program was instituted to generate longer-term stress relaxation data for several alloys commonly used in spacecraft springs. Because it would be impractical to test the myriad types and sizes of springs used for spacecraft hardware, a more general approach was taken. A database of stress relaxation behavior was generated by testing wires in tension, and an analytic method was developed to apply this database of uniform tensile behavior to the more complex, non-uniform stress states that exist within the cross section of springs.

Important material considerations affecting the stress relaxation behavior of metals are alloy composition, strength level, heat treatment, degree of cold work, and residual stresses. Significant test parameters to consider are the stress field, the initial stress level, time, and temperature. In this study, an effort was made to address all of these factors, either through material selection or test planning. However, it is worthwhile to note that residual stresses, which are the result of non-uniform plastic yielding of a

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spring cross section, cannot be generated in a uniform tensile stress field. For this reason, they were considered in the analytical, rather than the experimental portion of the study. An understanding of the existence and magnitude of residual stresses is needed for modeling, but experimental determination can only be made using the springs themselves.

Temperature excursions encountered during normal storage would have a slight effect on the stress relaxation behavior of spring alloys. However, maintaining a constant temperature during testing is important to eliminate complicating effects arising from differential thermal expansion between the sample and the testing apparatus. For this reason, all stress relaxation testing was done in a constant temperature environment, controlled to within $21^{\circ} \pm 1.5^{\circ}\text{C}$.

Materials

Five alloys were selected for testing in this program: 302 and 17-7 PH stainless steels; steel music wire; beryllium-copper alloy 25; and Elgiloy, a cobalt alloy. These are common alloys in general use for spacecraft springs and also represent a wide variety of compositions, strength levels, heat treatments, and cold-worked conditions. Details of the tested alloys are listed in Table 1.

The 17-7 PH stainless steel and Elgiloy are high-strength, precipitation-hardening alloys. Wire of these alloys intended for spring fabrication is supplied in the solution-annealed condition, requiring an age-hardening heat treatment to develop final properties of the solution treated and aged (STA) condition. Aging would normally be done after spring fabrication, but for the uniaxial wire tests, was done beforehand. The other alloys were supplied in their final conditions, requiring no additional heat treatment, except for a limited number of music wire and 302 stainless-steel samples that were tempered at either 260 or 366°C. For three of the alloys, two diameters of wire were tested in order to investigate the effect of cold work and strength level. Compared to larger-diameter wires, smaller wires require additional cold work during manufacture and are generally at higher strength levels.

Table 1. Alloys Tested for Stress Relaxation in Uniaxial Tension

Alloy	Specification	Tested Condition	Wire Diameter mm	Wire Diameter in
Be-Cu Alloy #25	ASTM B197-91	Tempered, 1/2H	0.51 & 1.27	0.02 & 0.05
Steel Music Wire	ASTM A228-91	As-Drawn	0.51 & 1.27	0.02 & 0.05
Steel Music Wire	ASTM A228-91	Drawn and Tempered	0.51	0.02
302 Stainless Steel	ASTM 313-91	As-Drawn	0.51 & 1.27	0.02 & 0.05
302 Stainless Steel	ASTM 313-91	Drawn and Tempered	0.51	0.02
17-7 PH Stainless Steel	AMS 5678B	Solution Treated & Aged	1.27	0.05
Elgiloy	AMS 5833	Solution Treated & Aged	1.27	0.05

Tensile properties were determined for 0.51 mm (0.02 in) diameter wires of Be-Cu, music wire, and 302 stainless steel, and for 1.27 mm (0.05 in) diameter wires of all of the alloys, by uniaxial tensile testing of three replicate samples. The results of these tests are given in Table 2, where F_{TU} is the ultimate tensile strength, F_{TY} is the 0.2% off-

set tensile yield strength, and F_{PL} is the tensile proportional limit stress of the wire.³ The proportional limit stress is a rough measure of the elastic limit strength of a material and has been defined in various ways. For present purposes, F_{PL} is taken to be the stress at 100 microstrain (0.01%) offset.

Stress relaxation of several compression springs of 17-7 PH and 302 stainless steel from actual space hardware were also tested during this study. This was done primarily to provide needed data for the springs, but it also allowed for comparison of the torsional stress data from the springs with the tension data from the wire tests.

Table 2. Tensile Properties of the Tested Alloys

Alloy	Condition	Wire Diameter mm (in)	F_{TU} MPa (ksi)	F_{TY} MPa (ksi)	F_{PL} MPa (ksi)
Music Wire	As Drawn	1.27 (0.05)	2270 (326)	1970 (282)	1150 (164)
		0.51 (0.02)	2750 (393)	2640 (378)	1730 (248)
Music Wire	Tempered 260 °C	0.51 (0.02)	2650 (379)	2620 (375)	2160 (310)
Music Wire	Tempered 366 °C	0.51 (0.02)	2310 (331)	2060 (295)	1690 (242)
Be-Cu Alloy 25	1/2 Hard	1.27 (0.05)	1330 (191)	1180 (170)	683 (98)
		0.51 (0.02)	1300 (186)	1220 (175)	712 (102)
302 SS	As Drawn	1.27 (0.05)	1930 (277)	1730 (247)	944 (135)
		0.51 (0.02)	2330 (333)	2020 (289)	1260 (181)
302 SS	Tempered 366 °C	0.51 (0.02)	2520 (361)	2280 (326)	1480 (213)
17-7 PH SS	STA	1.27 (0.05)	2190 (313)	2030 (291)	1190 (171)
Elgiloy	STA	1.27 (0.05)	2210 (317)	1960 (280)	955 (137)

Experimental Procedures for Wire Testing

The stress relaxation data were obtained at constant temperature by measuring the time-dependent decrease in stress in the wires while they were constrained at a constant displacement. All testing equipment and measurement procedures met or exceeded ASTM requirements for stress relaxation testing.⁴ The wires were mounted and held in one of twelve identical test fixtures located in the constant temperature facility. A schematic of the fixture is shown in Figure 1.

Each fixture consists of two square base plates separated by four cylindrical posts. There are two capstan-type gripping devices, one to hold each end of the wire. The upper grip is attached to a load cell, while the lower one is attached to the bottom base plate. The load cell is affixed to a threaded rod that passes through a hole in the center of the upper base plate. Load is applied to the wire by tightening a tensioning nut, which is screwed onto the threaded rod and bears against a precision washer atop the upper base plate.

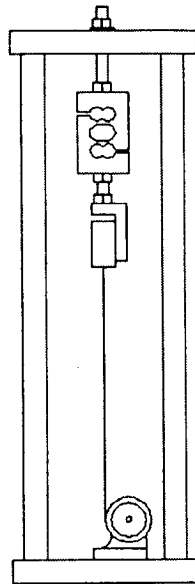


Figure 1. Fixture for stress relaxation testing of wire. A wire sample is shown between the capstan grips, which are rotated 90° with respect to each other here to provide a more detailed view.

To ensure that all of the stress relaxation occurs in the sample rather than the fixture, it is important that the fixture be rigid compared to the wire. Thus all the components were fabricated from steel. The cylindrical posts are 2.5 cm (1.0 in) in diameter, and the base plates are 2.5 cm (1.0 in) thick by 20.3 cm (8.0 in) on a side. The 1.3 cm (0.5 in) diameter bolts and threaded rods are used to attach the capstan grips and the load cell. The capstan grips are 5.1 cm (2.0 in) in diameter. The overall dimensions of the fixture are 91.4 cm (36.0 in) high by 20.3 cm (8.0 in) on a side. Nominal gauge length of the tested wires is 61 cm (24.0 in). Maximum force rating of the load cell is 2.2 kN (500 lb).

The tests are conducted by carefully attaching each end of the wire to one of the capstan grips, then wrapping three times around by rotating the capstan so as not to twist the wire. After tightening down the capstans, the tensioning nut is screwed down to apply just enough force on the wire to hold it in place. Next, the slack is removed from the wire wraps while maintaining the slight tension. The wire is then loaded to a pre-selected value and placed aside with no further disturbance. Load, time, and temperature are monitored continuously throughout the tests using analog strain-gauge signal conditioners and computer-controlled digital data logging equipment.

Stability of the load monitoring instrumentation is important when conducting long-term tests. To investigate stability of the equipment, a weight of 2.1 kN (483 lb) was applied to one of the 2.2 kN (500 lb) load cells for a period of almost 2,000 h. The results of that test are shown in Figure 2, where the stability of the reading is seen to be a near perfect 0.007% per log hour.

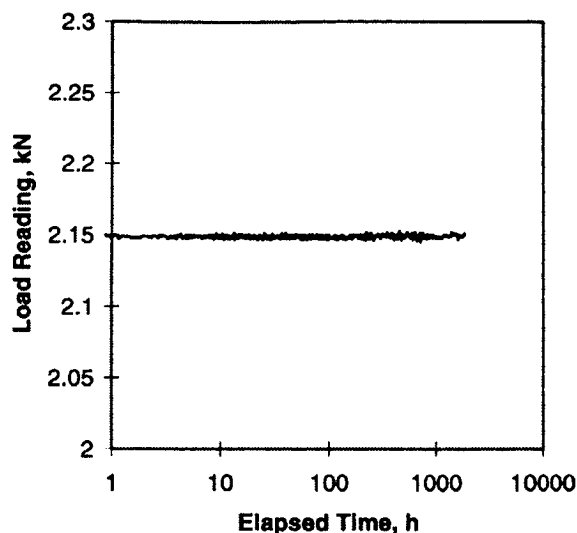


Figure 2. Stability of a 2.2 kN (500 lb) load cell and instrumentation with a hanging load. Error = 0.007 % per log hour.

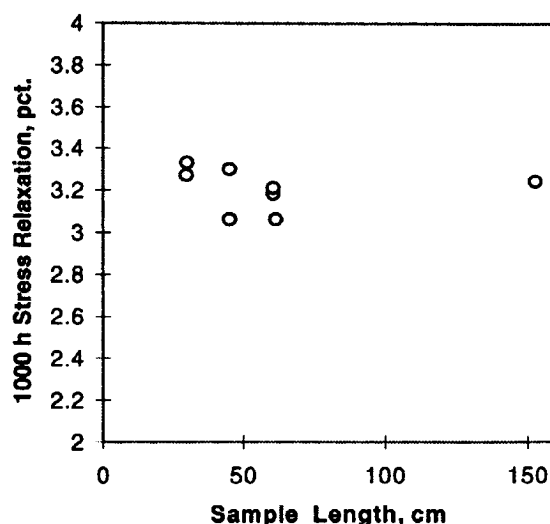


Figure 3. Stress relaxation after 1,000 hours for 1.27 mm (0.05 in) 302 SS samples. Nominal initial stress = $0.64 \pm 0.01 F_{TU}$.

A major concern was that some slippage of the wire on the grips would occur during the test. Any slippage would result in extraneous deformation of the wires, thus introducing errors into the data. Slippage would be independent of the wire length, and so would be expected to affect the stress relaxation data more severely for shorter wires. To settle this question, long-term tests were performed on eight wires of 302 stainless steel. Sample lengths were either 31, 46, 61, or 152 cm (8, 18, 24, or 60 in). The longest sample was accommodated by stringing it between two test fixtures that had been clamped tightly together, end to end. Nominal test durations were 8,000 h. Results of these tests are presented in Figure 3, where the percentage of stress relaxation after 1,000 h is plotted versus gauge length. The graph shows that there is no significant effect of sample length on the stress relaxation. This is strong evidence that no slippage is occurring in the grip ends of the wires and that the stress relaxation data obtained by using this fixture are valid.

Another possible source of error during long-term testing is equipment breakdown or power failure. During the course of this test program, which lasted approximately four years, there were several electrical power interruptions. Each time, the facility was restarted, and temperature stability was re-established to within $21^\circ \pm 1.5^\circ\text{C}$. In some cases, particularly for samples in which little time had been invested at that point, tests were simply aborted. On most occasions, data logging was resumed after restoration of the power with no harmful consequences. However, in a few instances where data gathering continued on existing samples, a small shift in the curve was noted. An example is shown in Figure 4a. Clearly, such an abrupt shift represents a slight

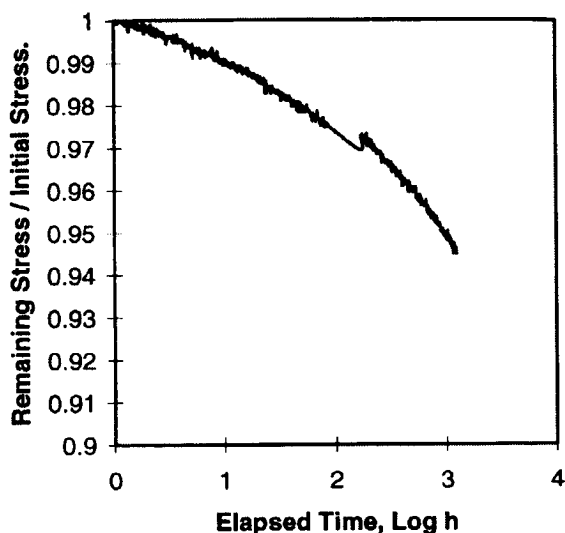


Figure 4a. Stress relaxation of a Be-Cu wire sample showing data shift upon restarting after a power outage.

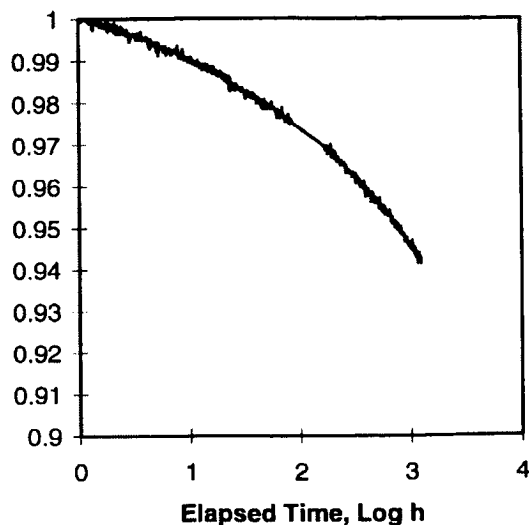


Figure 4b. Curve of Figure 4a after correction for the data shift.

change in the electrical balance point of the instrumentation, rather than an actual change in the stress relaxation behavior of the sample. Following this reasoning, these data shifts were corrected, as necessary, by superimposing a constant correction factor on the remaining portion of the data. The correction was made by curve-fitting the existing data, extrapolating to the restarting point, then applying the appropriate correction factor from that point onward. Figure 4b shows the correction of the data of Figure 4a.

Experimental Results of Wire Testing

A total of 94 wire samples was studied for periods ranging from 1,000 to 10,000 h. Representative stress relaxation data for the different alloys are plotted in Figure 5, where the fractional reduction of initial stress is plotted versus the logarithm of elapsed time. Because the initial stress at time zero is undefined for a logarithmic plot, a suitable value had to be selected. Although stress reduction begins immediately upon loading, the initial decline is somewhat variable, depending on loading rate. For all samples tested, this transient behavior, as observed on a semi-logarithmic plot, was completely attenuated after approximately 0.5 h. Therefore, for purposes of this study, initial stress is defined as the stress remaining after 1 h of testing.

The slope of a straight line drawn from the origin through the 1,000-h point in each curve of Figure 5 would represent the 1,000-h stress relaxation rate. The curves of Figure 5 show that the wires of all alloys, except music wire and the Be-Cu alloy,

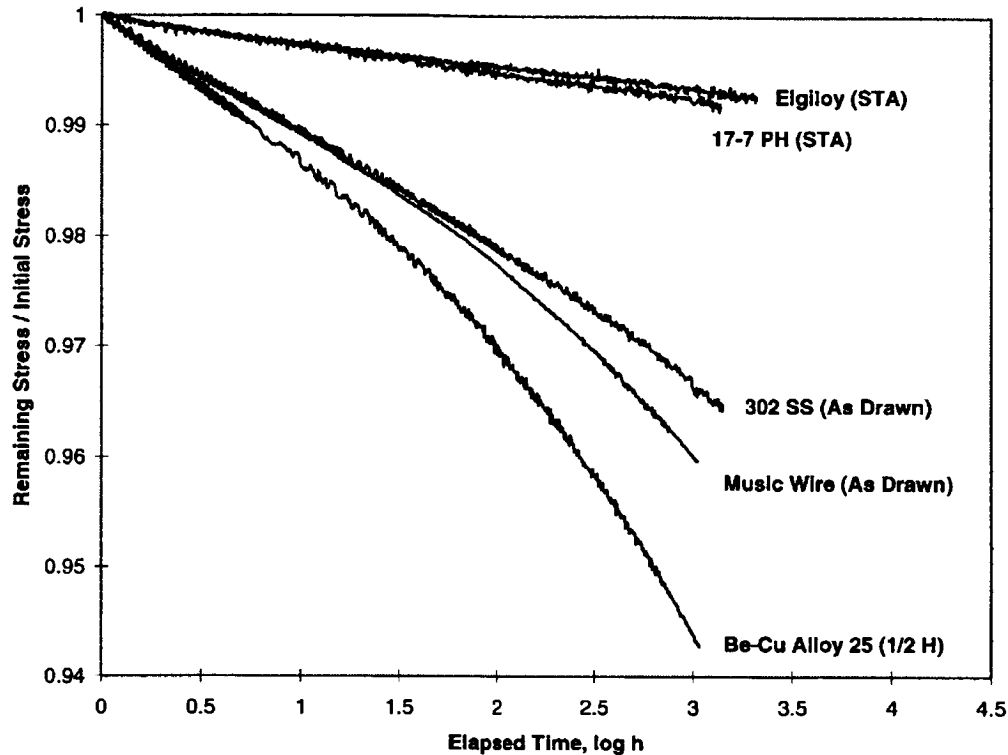


Figure 5. Stress relaxation of five 1.27 mm (0.05 in) diameter, high-strength spring alloy wires. Nominal initial stress = $0.66 \pm 0.02 F_{TU}$.

exhibit reasonably linear behavior on this semi-logarithmic plot, and so can be characterized by this rate. Analysis of long-term (8,000 h) 302 stainless-steel data has revealed that the slight nonlinearity evident in these data results in an insignificant error (less than 1%) when extrapolated from 1,000 to 8,000 h. Further linear extrapolation to periods of up to three years (approximately 26,000 h) is reasonable. Because of the convex curvature of Be-Cu and music wire data, linear extrapolation beyond 1,000 h may not be warranted. This is especially noteworthy since the extrapolation would be non-conservative, which illustrates the danger of linear extrapolation of short-term (≤ 100 h) stress relaxation data to predict stress relaxation during long-term spacecraft storage.

A number of mathematical functions were investigated in an effort to establish a viable technique to extrapolate the nonlinear music wire and Be-Cu data. Empirical curve fitting by a third-order polynomial was found to provide an excellent correlation, as illustrated in Figure 6, where the Be-Cu data of Figure 5 are re-plotted. In this figure, only the data up to 1,000 h were used for the curve fit. The curve beyond 3 log hours represents an extrapolation. It is clear that for these data, the polynomial representation is much more conservative than the linear one, and turns out to be surprisingly accurate in predicting the stress relaxation behavior beyond 1,000 h. Although not shown here, this technique produces excellent extrapolations for the nonlinear music

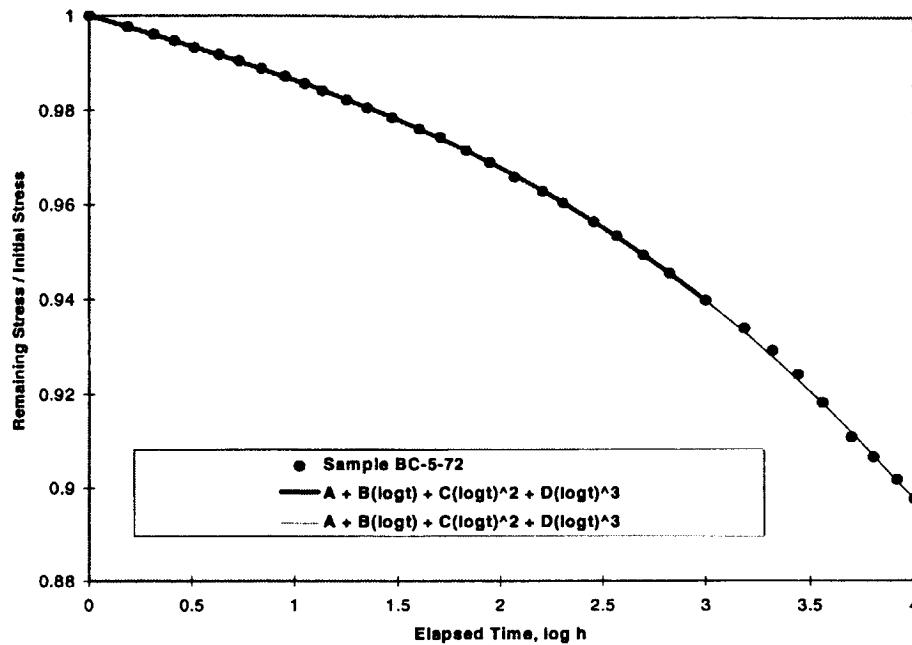


Figure 6. Third-order polynomial curve-fit of data for the Be-Cu wire sample of Figure 5. Only the data from 0 to 3 log hours were used for the analysis. The curve beyond 3 log hours is an extrapolation.

wire data as well. There is no known physical justification for this polynomial representation, however. It is presented here as an empirical finding only.

A few of the 0.51 mm (0.02 in) diameter music wire and 302 stainless-steel samples were tempered before being tested to investigate possible beneficial effects on the stress relaxation behavior. Figure 7 shows the results of those tests. In each case, it is seen that tempering does reduce the stress relaxation rate compared to the as-drawn wire, probably by rearranging the dislocation structure into a more favorable state. This may also be the case for 17-7 PH and Elgiloy, whose aging treatments could well provide a similar annealing effect. The stress relaxation behavior of these alloys is far superior to the other alloys tested, as evident in Figure 5.

Data from the 0.51 mm (0.02 in) wires of Figure 7 appear noisier than those from the 1.27 mm (0.05 in) diameter wires of Figure 5. The noise is due to minor temperature fluctuations of the constant temperature laboratory ($\pm 1.5^\circ\text{C}$), which cause differential expansion between the fixture and the sample. It is more noticeable for the smaller wires because the lower loads from these tests require higher amplification. The noise does not affect the outcome of the tests because it tends to average out.

A summary of the stress relaxation after 1,000 hours as a function of initial stress ratio for all 94 samples tested in this study is shown in Figure 8. Most of the data are from 1.27 mm (0.05 in) diameter wires, but about 50% of the music wire, 302 stainless steel,

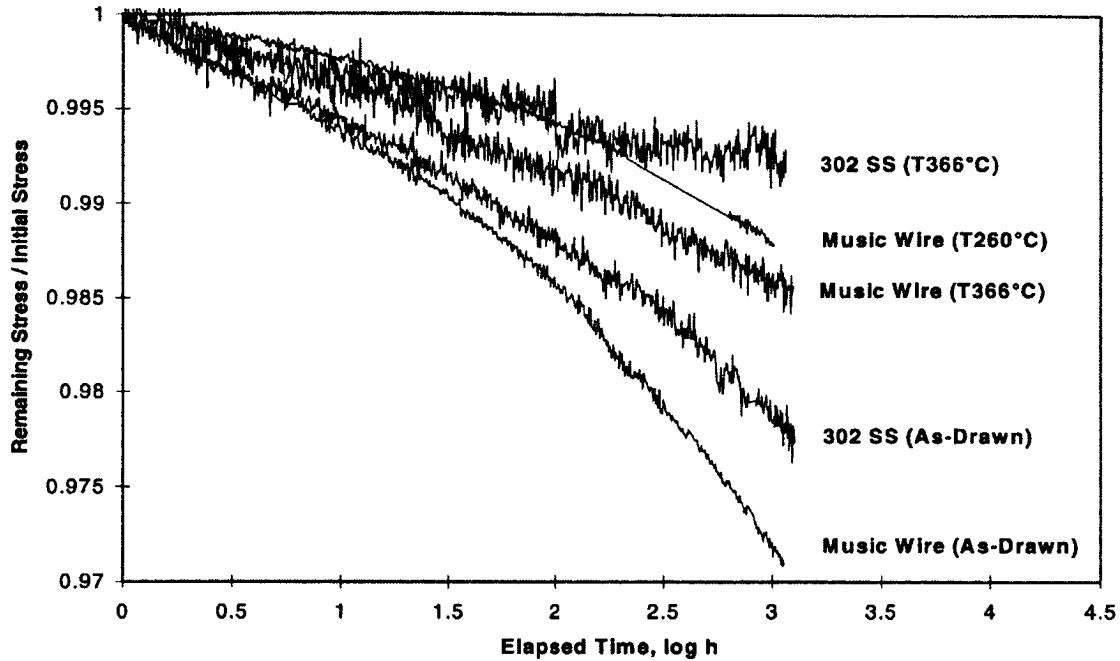


Figure 7. Data curves showing the effect of tempering on the stress relaxation behavior of 0.51 mm (0.02 in) diameter music wire and 302 stainless steel samples. Nominal initial stress = $0.44 \pm 0.03 F_{TU}$.

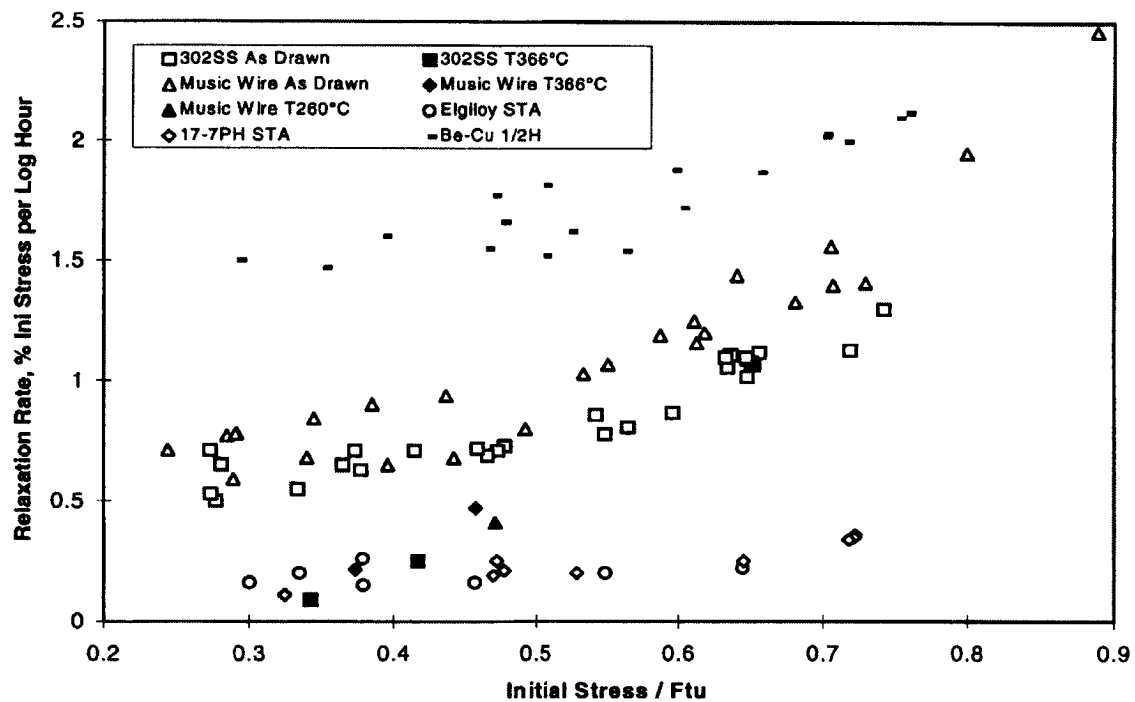


Figure 8. Stress relaxation after 1000 hours for five high-strength spring alloys.

and Be-Cu tests were done on 0.51 mm (0.02 in) diameter wire. The data from both sample sizes are intermixed in Figure 8 because there is no discernible effect of wire size on the results.

Figure 8 shows again that Elgiloy and 17-7 PH stainless steel are highly resistant to stress relaxation compared to the other alloys, although the limited data of the tempered music wire and 302 stainless steel appear nearly comparable. The stress relaxation rate of Be-Cu alloy 25 is almost an order of magnitude higher than the best materials here. Another interesting observation from Figure 8 is that, at high values of initial stress, the stress relaxation rate increases rapidly as the initial stress ratio increases. However, below a stress ratio of approximately 0.5, the rate tends to become constant. Since the conservative design of spacecraft hardware generally calls for stresses in this lower regime, analysis of stress relaxation in spacecraft springs can be greatly simplified by assuming the stress relaxation rate, expressed as a ratio of the initial stress, to be independent of the initial stress level.

Analytical Modeling

Basic Theories

Tension/compression and torsional springs are the two most common springs used in storing energy. Tension/compression springs are normally made from constant cross-sectional wires wound in helical shapes. Depending upon the direction of the applied force, the overall length of the spring will either extend or contract. Because of the fabrication of these springs, the wire itself is subject to a combination of shear force and torsion about the center of the cross-section. The applied torsion is the dominate stress source. Torsion springs, on the other hand, are designed to store energy generated from torsional moment. The wire is under axial force with bending as the primary source of stress. Moreover, either type of stress is non-uniform. Because of the stress non-uniformity, the stress relaxation of the spring assembly is difficult to assess experimentally. Therefore, an analytical prediction for the spring assembly is necessary.

In attempting to apply the tension stress-relaxation data to shear stressed conditions, a shear stress-strain curve from a tensile stress-strain curve was constructed based on the following two facts:⁵

1. Plastic strain is caused solely by shear strain;
2. Maximum shear stress is one-half of the tensile stress in a uniaxial tensile specimen.

For simplicity, it is assumed that the uniaxial stress-strain curve is bilinear, as shown in Figure 9a. The modulus of elasticity is E , and the elastic limit stress and strain are σ_0 .

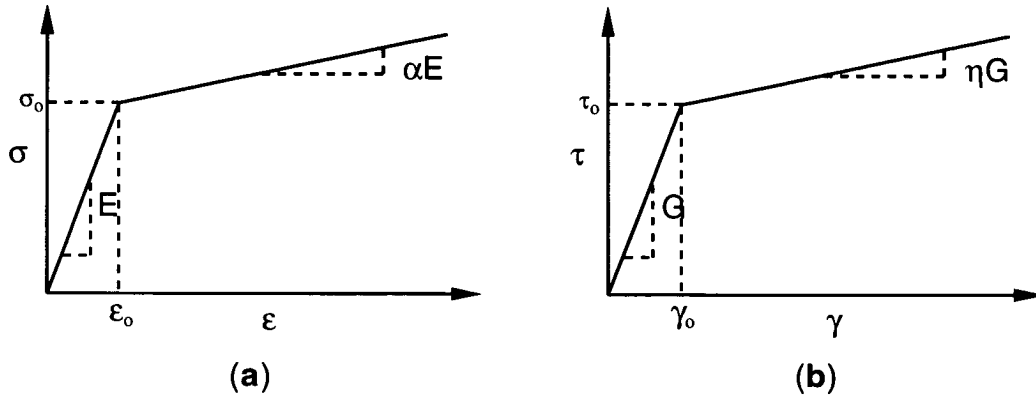


Figure 9. Uniaxial and shear bilinear stress-strain curves.
(a) Bilinear uniaxial tensile stress-strain curve;
(b) bilinear shear stress-strain curve.

and ϵ_0 , respectively. The modulus in the inelastic portion of the curve is αE . Similarly, a shear stress-strain curve is constructed as shown in Figure 9b. The elastic limit stress and strain are τ_0 and γ_0 , respectively. The shear modulus in the elastic region is G , and that in the inelastic region is ηG . We shall establish the value of η in terms of α and other material constants.

Let us define (σ, ϵ) and (τ, γ) to be two points on the tensile and shear stress-strain curves, respectively. The slope of the inelastic portion of the tensile curve is

$$(\sigma - \sigma_0)/(\epsilon - \epsilon_0) = \alpha E. \quad (1)$$

Similarly, the slope of the inelastic portion of the shear curve is

$$(\tau - \tau_0)/(\gamma - \gamma_0) = \eta G. \quad (2)$$

From fact 2, above,

$$\begin{aligned} \tau &= \sigma/2, & \tau_0 &= \sigma_0/2 \\ \gamma &= 3\epsilon/2, & \gamma_0 &= \tau_0/G, \end{aligned}$$

and taking into account the fact that Poisson's ratio in the inelastic region is $1/2$, we have

$$\begin{aligned} (\tau - \tau_0)/(\gamma - \gamma_0) &= [(\sigma - \sigma_0)/2]/(3\epsilon/2 - \tau_0/G) \approx [(\sigma - \sigma_0)/2]/(3\epsilon/2 - 3\epsilon_0/2) \\ &= \alpha E/3. \end{aligned} \quad (3)$$

This means that the ratio of the inelastic shear modulus to elastic shear modulus in a shear stress-strain curve is equal to the ratio of the inelastic modulus to elastic Young's modulus in a uniaxial stress-strain curve.

Relation Between Tensile and Shear Stress Relaxation

Let $F(\sigma_0, t, M, T)$ and $G(\tau_0, t, M, T)$ be the stress-relaxation functions associated with tension and shear modes, respectively. Then we have

$$\sigma = \sigma_0 F(\sigma_0, t, M, T) \text{ for tension} \quad (4)$$

and

$$\tau = \tau_0 G(\tau_0, t, M, T) \text{ for shear,} \quad (5)$$

where σ and τ are the stresses at time t , σ_0 and τ_0 are the initially applied stresses, M represents the materials and specimen structure parameters, and T is the temperature. Based on facts 1 and 2 in the Basic Theory section, together with the fact that part of the elastic strain is converted into plastic strain during stress relaxation, it is concluded that

$$G(2\tau_0, t, M, T) = F(\sigma_0, t, M, T), \quad (6)$$

and that the uniaxial tension stress-relaxation data can be applied to shear loaded cases.

Relaxation Test Data for 302 Stainless Steel

Data from a 302 stainless-steel V-band spring can be used to validate the model. The relaxation function $F(\sigma_0, t, M, T)$ for this case is independent of M and T . From the uniaxial test data, the tension relaxation function was expressed as

$$F(\sigma_0, t, M, T) = 1 - S \log(t), \quad (7)$$

where t is in hours. The S is expressed as a linear function of the applied initial stress, S_0 , i.e.,

$$S = S_0 + \kappa \sigma_0.$$

$$\sigma_0 = 903 \text{ MPa (131 ksi), } S = 0.0065$$

and $\sigma_0 = 1234 \text{ MPa (179 ksi), } S = 0.0116$,

gives $S_0 = -0.0074$, and $\kappa = 1.537 \times 10^{-8} ((\text{MPa})^{-1})(0.000106(\text{ksi})^{-1})$.

The above-established values for S_0 and κ , as well as Eq. (7), were applied to a 302 stainless-steel tension/compression spring. The spring is used in a satellite program. The dimensions of the spring, the applied initial load, the moment, and the load relaxation are listed as follows:

Spring diam	= 19.76 mm (0.778 in)	Wire diam	= 1.78 mm (0.07 in)
Initial load	= 72.1 N (16.2 lb)	Final load	= 70.5 N (15.86 lb)
Test time	= 1000 h (extrapolated)	Percentage of load relaxation	= 2.1%.

We now predict the torque relaxation in a compression spring by the use of Eq. (6) and compare the prediction with the actual experimental observation. Consider a compression spring of circular cross-sectioned wires. We designate:

M = the applied torque equal to $PD/2$; P = applied compression load;
 D = spring diam; and d = wire diam.

The shear stress in the wire is expressed as

$$\tau = (16PD r / \pi d^4) K, \quad (8)$$

with a maximum value of

$$\tau_{\max} = (8PD / \pi d^3) K,$$

where r is the radial location from center of the wire where the shear stress is calculated, K is a stress concentration factor equal to $(4C - 1)/(4C - 4) + 0.615/C$, and C is equal to D/d .

To determine the torque relaxation in the spring, we use Eqs. (6), (7), and (8) in the following torque equation for the wire cross section:

$$\tau / \tau_0 = 1 - (S_0 + 2\kappa \tau_0) \log(t) \quad (9)$$

$$M_t = 2\pi \int_0^{d/2} \tau_0 r^2 dr$$

which gives

$$M_t / M_0 = 1 - S_0 \log(t) - (128M_0 / 5\pi d^3) \kappa \log(t), \quad (10)$$

where

$$M_0 = \tau_{\max} \pi d^3 / 16PD.$$

A substitution of the S_0 and κ values into Eq. (10) yields a M_t / M_0 ratio of 0.974. In other words, the predicted load relaxation for this compression spring is 2.6%, which compares well with the 2.1% measured value. The difference is due to several possible reasons, including the bilinear stress-strain and linear S vs s_0 assumptions used in the derivation.

Determination of Residual Stresses

Residual stresses are often generated in both compression and torsion springs by presetting the springs. The residual stresses occur as a result of the plastic yielding of the spring wires, followed by an unloading of the spring. The residual stresses always act in the opposite direction to the operating stresses, thus increasing the range of the operating stress without inducing further plastic yielding. As will be seen from the calculation, the residual stresses can be a significant portion of the yield strength and

may experience a relaxation themselves. The determination of the residual stresses becomes an important portion of the overall relaxation prediction.

Residual Stresses in a Compression Spring with a Circular Wire Cross Section

Assume that part of the wire cross section is plastically yielded due to the application of an external torque M_t , and that the stress distribution in the wire is bilinear, as shown in Figure 9b. At a distance aR from the center of the wire, the shear stress is τ_0 .

Therefore, the stress, τ , as a function of r is expressed as

$$\tau = \tau_0 r / (aR) \quad 0 \leq r \leq aR; \quad \tau = [(1 - \alpha) + \alpha r / (aR)] \tau_0 \quad aR \leq r \leq R. \quad (11)$$

The external M_t is expressed as

$$M_t / M_e = [a^3 + 4(1 - \alpha)(1 - a^3)/3 + \alpha(1 - a^4)/a], \quad (12)$$

where $M_e = \pi \tau_0 R^3 / 2$ is the torque with a maximum shear stress equal to τ_0 at $r = R$. For 302 stainless-steel wire, the value of τ_0 is 741 MPa (107.5 ksi). For a given value of M_t / M_e , the corresponding value of a can be solved algebraically. The shear stress distribution in the wire can be then determined using Eq. (11). The residual stress following an elastic unloading is expressed as

$$\tau' = -2 M_t r / \pi R^4. \quad (13)$$

A combination of Eqs. (11) and (13) gives the distribution of the shear stresses for any given M_t / M_e . Figure 10 depicts the residual stress distribution for 302 stainless steel with a 1.27 mm (0.05 in) wire diameter and a M_t / M_e value of 2. The maximum residual stress is about 165 MPa (24 ksi), which is approximately 22% of the elastic limit stress of 302 stainless steel in shear.

Residual Stress Determination for a Torsion Spring with a Rectangular Wire Cross Section

Assume the wire has a width b (in the direction parallel to the axis of the spring) and a depth of $2D$ (in the direction parallel to the radius of the spring). Again, assume the cross section is partly yielded plastically due to the application of an external bending moment, M_b . The bilinear stress-strain curve is as shown in Figure 9a. The elastic stress, σ_0 , is 1482 MPa (215 ksi). Similar to the distributions in the shear, the elastic and inelastic stress distributions in the cross section are expressed as

$$\sigma = \sigma_0 x / (aD) \quad 0 \leq x \leq aD; \quad \sigma = [(1 - \alpha) + \alpha x / (aD)] \sigma_0 \quad aD \leq x \leq D. \quad (14)$$

The bending moment, M_b , is

$$M_b / M_{be} = [a^2 + 3(1 - \alpha)(1 - a^2)/2 + (1 - a^3)\alpha/a] \quad (15)$$

where

$$M_{be} = 2\sigma_0 b D^2 / 3.$$

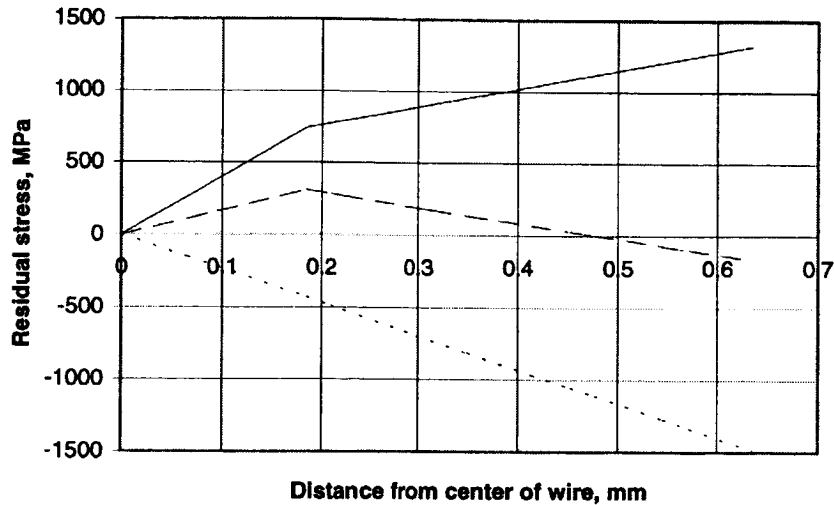


Figure 10. Residual shear stress for 302 stainless steel 1.27 mm (0.05 in) diameter wire (M_b/M_{be} ratio = 2.0). Solid line is nonlinear loading; dotted line, linear unloading; dash-dotted line, residual stress.

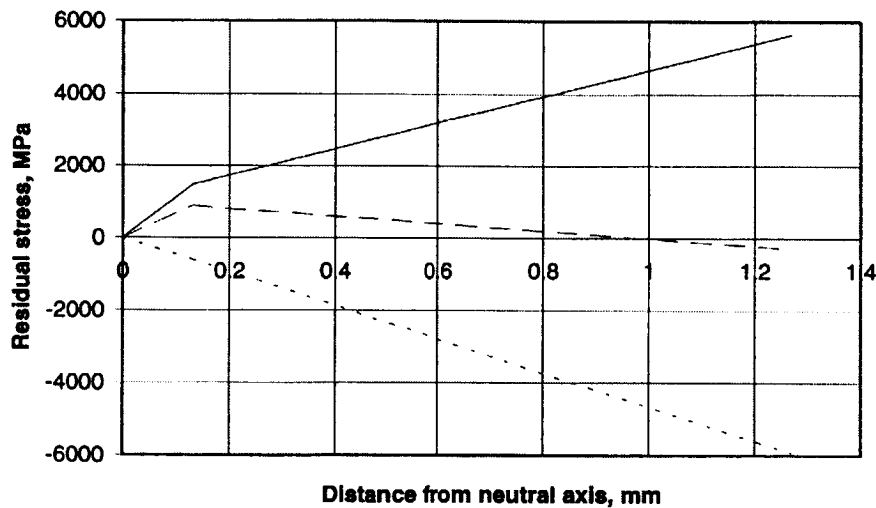


Figure 11. Residual bending stress for 302 stainless steel 1.27 mm (0.05 in) wide and 2.54 mm (0.10 in) rectangular cross sectioned wire (M_b/M_{be} ratio = 2.0). Solid line is nonlinear loading; dotted line, linear unloading; dashed line, residual stress.

For unloading elastically, the stress distribution for the moment, M_b , is expressed as

$$\sigma_o' = -3M_b x / (2bD^3). \quad (16)$$

A combination of Eqs. (14) and (16) gives the residual stresses in the cross section. A case with M_b/M_{be} of 2 was calculated for $b = D = 1.27$ mm (0.05 in). The residual stress distribution is depicted in Figure 11. A residual compressive stress of 296 MPa (43 ksi) corresponding to 20% of the elastic limit stress is obtained.

Summary

Stress relaxation testing results for 0.51 and 1.27 mm (0.02 and 0.05 in) wires of 302 stainless steel, music wire, and Be-Cu, and for 1.27 mm (0.05 in) wires of Elgiloy and 17-7 PH stainless can be used for predicting the stress relaxation behavior of springs made from these alloys. Linear extrapolation of the 1000-h data has been shown to be reasonable for the 302 stainless steel, 17-7 PH stainless steel, and Elgiloy. It was shown that the high-strength, precipitation-hardening alloys, 17-7 PH stainless steel and Elgiloy, are by far the best choice for alloy selection when stress relaxation of springs is a major concern. However, tempering of cold-worked music wire and 302 stainless steel imparted significant improvements and stress relaxation resistance of these materials.

An analytical technique was developed to predict the stress relaxation for compression and torsion springs based on the uniaxial wire data. In this technique, the tensile stress relaxation function for 302 stainless-steel wires has been used successfully to predict the torque reduction for a 302 stainless-steel V-band compression spring. The predicted 2.6% load reduction after 1000 h compares well with the 2.1% load reduction obtained from tests.

Residual stress prediction for both compression and torsion indicate that the residual stresses can be significant portions of the elastic-limit stress and are beneficial in actual satellite or launch operations.

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